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HOCHSCHULE HEILBRONN

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Notation and Formulae Statistics

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1 Notation (commonly used in Textbooks)

1.1 Population and Sample Elements

Symbol	Meaning
$X, Y, Z, \ldots$(capital letters)	Random variables (range = population elements)
$x, y, z, \ldots$(lowercase letters)	A subset of the range of a random variable (e.g., a sample)
$x_i, y_i, z_i, \ldots$(lowercase letters with subscript)	A single specific value (number) that a random variable may take on
N	Population size
n	Sample size

1.2 Probabilities

Symbol	Meaning
$P(x)$	Short for $P(X \in x)$, i.e., the probability that event x occurs. Examples: • Random variable takes on x_i $(x = x_i)$: $P(x_i) \coloneqq P(X = x_i)$ • Random variable in interval $x = [x_i, x_j]$: $P([x_i, x_j]) \coloneqq P(x_i \leq X \leq x_j)$
$P(x \mid y)$	Short for $P(X \in x \mid Y \in y)$, i.e., the conditional probability that event x occurs, given that event y has occurred.
$P(x, y)$	Short for $P(X \in x, Y \in y)$, i.e., the joint probability that x and y occur.
f_X	Probability densitiy function (pdf) of a continuous random variable X
F_X	Cumulative distribution function (cdf) of a continuous random variable X

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1.3 Parameters and Statistics

Paramter / Statistic	Notation Parameter	Notation Statistic
Population / sample *mean*	μ	$\bar{x}$
Population / sample *standard deviation and variance*	$\sigma \; or \; sd(X)$ $\sigma^2 \; or \; var(X)$	$s \; or \; sd(x)$ $s^2 \; or \; var(x)$
Population / sample *covariance* of the variables X and Y	$\sigma_{XY} \; or \; cov(X,Y)$	$s_{xy} \; or \; cov(x,y)$
Population / sample *Pearson correlation coefficient* of the variables X and Y	ρ_{XY}	r_{xy}
Population / sample *Pearson partial correlation coefficient* of the variables X and Y when controlling for Z	$\rho_{XY\|Z}$	$r_{xy\|z}$
Population / sample Spearman rho correlation coefficient	ρ_{XY}^{S}	r_{xy}^{S}

1.4 Special Symbols

Symbol	Meaning
$\bar{X}$	Random variable of the sample mean
$E[X], E[x]$	Expected value of population / sample
se	Standard error (= standard deviation) of the distribution (random variable) of a statistic
$\hat{\mu}$	Estimate of the population mean
$\hat{\sigma}^2$	Estimate of the population variance
H_0, H_1	Null / Alternative Hypothesis

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2 Formulae for Variance, Covariance and Correlation

2.1 Expected Value

Let x be a sample of size n with v unique values, then:

$$E[x] = \frac{frequency\ x_1}{n} \cdot x_1 + \cdots + \frac{frequency\ x_v}{n} \cdot x_v = P(x_1) \cdot x_1 + \cdots + P(x_v) \cdot x_v,$$

i.e., the expected value is the probability-weighted sum of the unique values of a (sample) random variable.

2.2 Variance and Covariance

Let (x, y) be a sample of size n, then:

$$var(x) = \frac{(x_1 - \bar{x})^2 + \cdots + (x_n - \bar{x})^2}{n} = E[(x - \bar{x})^2]$$

$$cov(x, y) = \frac{(x_1 - \bar{x})(y_1 - \bar{y}) + \cdots + (x_n - \bar{x})(y_n - \bar{y})}{n} = E[(x - \bar{x})(y - \bar{y})]$$

Observe that $cov(x, x) = var(x)$.

2.3 Pearson Correlation Coefficient

Sample Pearson correlation coefficient:

$$r_{xy} = \frac{cov(x, y)}{sd(x) \cdot sd(y)}$$

2.4 Pearson Partial Correlation Coefficient

Sample Pearson partial correlation coefficient for *one contol variable* (z):

$$r_{xy|z} = \frac{r_{xy} - r_{xz} \cdot r_{yz}}{\sqrt{1 - r_{xz}^2} \cdot \sqrt{1 - r_{yz}^2}}$$

Sample Pearson partial correlation coefficient for k contol variables ($z_1, \ldots, z_k$):

$$r_{xy|z_1 \ldots z_k} = \frac{r_{xy} - (r_{xz_1} \cdot r_{yz_1} + \cdots + r_{xz_1} \cdot r_{yz_k})}{\sqrt{1 - (r_{xz_1}^2 + \cdots + r_{xz_k}^2)} \cdot \sqrt{1 - (r_{yz_1}^2 + \cdots + r_{yz_k}^2)}}$$

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3 Formulae for Conditional Probability, Independence, and Standard Score

3.1 Conditional Probability

The probability that x occurs, given y has occured is:

$$P(x \mid y) = \frac{P(y \mid x) \cdot P(x)}{P(y)} = \frac{P(x, y)}{P(y)}$$

3.2 Stochastic Independence

Two random variables are stochastically independent if

$$P(x, y) = P(x) \cdot P(y) \; for \; all \; x, y$$

3.3 Standard Score (z Score)

The z score of a value x_i of a random variable X is:

$$z(x_i) = \frac{x_i - mean_X}{sd(X)}$$

If X is normally distributed Z_X is standard normally distributed.

4 Formulae for Inferential Statistics

4.1 Finite Sample Correction of Standard Error

If

(i). sampling is done without replacement from a finite population and

(ii). the sample size (n) is large relative to the population size (N),

the standard error (se) must be muliplied by fpc (finite population correction):

$$se = se_{without\ fpc} \cdot fpc,$$

where $fpc = \sqrt{\frac{N-n}{N-1}}$.

4.2 Point Estimators

- The estimator for the population mean is the sample mean:

$$\hat{\mu} = \bar{x}$$

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- The estimator for the population variance is the Bessel-corrected sample variance:

$$\hat{\sigma}^2 = s^2 \cdot \frac{n}{n-1}$$

- The estimator for a correlation coefficient is the sample correlation coefficient:

$$\hat{\rho} = r$$

4.3 Interval Estimators

- The general formula for an interval estimator is:

$$point\ estimate \pm moe,$$

  where the margin of error (moe) is some multiple of the standard error (se):

$$moe = m \times se$$

- Interval estimator of mean with confidence level α and known population variance:

$$\bar{x} \pm z_{(1-\alpha)} \cdot se$$

4.4 P-Value for z Tests

Let ts be the test statistic, then

$$p = 1 - \Phi(|ts|) = P(X > |ts|),$$

where Φ is the cumulative distribution function of the standard normal distribution. Then:

$$One - tailed\ p - value = p$$

$$Two - tailed\ p - value = p \times 2$$

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5 Test Statistics of Parametric Tests

5.1 One- and Two-Sample Z Test of Means

5.1.1 One-sample z test of means

Let x be a sample of size n (with sample mean $\bar{x}$) of random variable X (with poulation standard deviation σ).

- The $standard\ error\ (se)$ of the distribution of sample means ($\bar{X}$) is:

$$se = \frac{\sigma}{\sqrt{n}}$$

- The $test\ statistic$ of testing X against a pre-specified level μ_0 is:

$$z = \frac{\bar{x} - \mu_0}{se}$$

5.1.2 Two-sample z test of means

Let x, y be samples of size n_x, n_y (with sample means $\bar{x}, \bar{y}$) of random variable X, Y (with poulation standard deviation σ_X, σ_Y).

Then, the random variable $D = X - Y$ has sample mean $\bar{d} = \bar{x} - \bar{y}$.

- The $pooled\ standard\ error\ (se)$ of the distribution of sample means ($\bar{D}$) is:

$$se_{\bar{D}} = \sqrt{se_X^2 + se_Y^2},$$

  where $se_X = \sigma_X/\sqrt{n_x}$ and $se_Y = \sigma_Y/\sqrt{n_y}$.

- The $test\ statistic$ of testing D against a pre-specified level μ_0 (= 0, usually) is:

$$z = \frac{\bar{d} - \mu_0}{se_D}$$

5.2 Correlation Tests

5.2.1 Pearson Correlation Test

Let r_{xy} be the sample correlation coefficient of a sample (x, y) of size n.

- The $standard\ error\ (se_R)$ of the distribution of correlation coeffients (R) is

$$se_R = \sqrt{\frac{1 - r_{xy}^2}{n - 2}}.$$

- The $test\ statistic$ of R against a pre-specified level ρ_0 (= 0, usually) is:

$$t_{n-2} = \frac{r_{xy} - \rho_0}{se_R},$$

  which follows a t-distribution with n-2 degrees of freedom.

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5.2.2 Pearson Partial Correlation Test

Let $r_{xy|z_1 \dots z_k}$ be the sample partial correlation coefficient of a sample $(x, y, z_1, \dots, z_k)$ of size n.

- The $standard\ error\ (se_R)$ of the distribution of correlation coeffients (R) is

$$se_R = \sqrt{\frac{1 - r_{xy}^2}{n - k - 3}},$$

  where k denotes the number of control variables.

- The $test\ statistic$ of R against a pre-specified level ρ_0 (= 0, usually) is:

$$t_{n-k-3} = \frac{r_{xy|z_1 \dots z_k} - \rho_0}{se_R},$$

  which follows a t-distribution with $n - k - 3$ degrees of freedom.

5.2.3 Fisher Transformation

- The $Fisher\ transformation$ of a correlation coefficient r is:

$$r^f = \frac{1}{2} \ln\left(\frac{1 + r}{1 - r}\right) = \operatorname{artanh}(r)$$

  and has the following standard error (se_f):

$$se_f = \sqrt{\frac{1}{n - 3}}$$

- The Fisher test statistic is then:

$$z_{fisher} = \frac{r_f}{se_f}$$

  and follows a standard normal distribution if n is sufficiently large.

A Fisher z Test based on the Fisher test statistic can be used to test any of the aforementioned correlations plus Spearman rank correlation.

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6 Test Statistics of Non-Parametric Tests

6.1 Mann Whitney U Test

- The U statistic is:

$$U = n_{max} \cdot n_{min} + \frac{n_{max}(n_{max}+1)}{2} - R_{max},$$

  where n_{max}, n_{min} are the sample sizes of the samples with the highest / lowest rank sum and R_{max} ist he highest rank sum.

- If there are no tied ranks, the standard error of U is:

$$se_u \sqrt{\frac{n_{max} \cdot n_{min}(n_{max}+n_{min}+1)}{12}}.$$

- The normal approximated $test\ statistic$ is:

$$z_U = \frac{U - \frac{n_{max} \times n_{min}}{2}}{se_U}.$$

6.2 Chi2 Test

- The chi^2 statistic is:

$$\chi^2 = \sum \frac{(O-E)^2}{E},$$

  where O are the observed and E are the expected frequencies.

- The normal approximated $test\ statistic$ is:

$$z_{\chi^2} = \frac{\chi^2 - dof}{\sqrt{2 \times dof}},$$

  where dof = degrees of freedom.

7 χ^2 − and T-Distribution

Let $Z_1, \dots, Z_k$ be independent standard normally distributed random variables, then the sum of their squares follows a χ^2 −distribution with k degrees of freedom:

$$\chi_k^2 = Z_1^2 + \cdots + Z_k^2.$$

Let Z be a standard normally distributed and χ_k^2 a χ^2 −distributed random variable, then the following is t-distributed with k degrees of freedom:

$$T_k = \frac{Z}{\sqrt{\chi_k^2/k}}.$$

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